INCH-POUND

MIL-P-70526A (AR)
6 July 1994
SUPERSEDING
MIL-P-70526 (AR)
29 October 1990

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, M913/XM927 DELAY ASSEMBLY FOR

This specification is approved for use by U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and quality assurance provisions for the fabrication and packaging of parts and assembly of the delay assembly for the 105MM, High Explosive, Rocket Assisted, M913/XM927 Projectiles. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Piactinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC **1315**

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SPECIFICATIONS

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing
MIL-P-217	-	Potassium Perchlorate (for pyrotechnics)
MIL-M-382	-	Magnesium Powder (for use in ammunition)
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-T-48140	-	Tungsten Powders (for pyrotechnics)
MIL-V-50433	-	Vinyl Alcohol - Acetate Resin

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-410	-	Non-destructive Testing Personnel Qualification and Certification
MIL-STD-00453	-	Inspection, Radiographic
MIL-STD-1234	-	Pyrotechnics; Sampling, Inspection and Testing

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.13)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT DRAWINGS

12926809 - Delay Assembly

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Procedures must be approved by the QPA.

INSPECTION EQUIPMENT DRAWINGS

9246401	- Nitrogen Test Assembly (reference only)
9246420	- Envelope Drawing (Porosity Test) (reference only)

CODE OF FEDERAL REGULATIONS

TITLE 49	- Transportation parts 100 - 199

(Copies other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by Contracting activity)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TEST AND MATERIALS

B430	- Particle size distribution of refractory metal-type powders by turbidimetry.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all the requirements specified on Drawing 12926809, all associated drawings, and with all requirements specified in applicable specifications and drawings.

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3.3 Pyrotechnic composition. The flash, delay and igniter compositions shall comply with applicable drawing requirements.

3.3.1 Total volatiles. The total volatiles of the flash, delay and igniter composition at the time of loading shall not exceed 0.10 percent when tested in accordance with 4.4.3.2.

3.4 Crimp strength. The washer crimp shall be capable of withstanding a 500 pound minimum load when tested as specified in 4.4.3.6.

3.5 Pyrotechnic porosity test. The pressure recorded at the exit end of the delay assembly, 80 seconds after application of the nitrogen gas, shall not exceed 9 PSIG above the zero reference. The pressure recorded at the exit end of the delay assembly 110 seconds after application of the nitrogen gas shall be no greater than 19 PSIG above the zero reference. The test shall be conducted in accordance with 4.4.3.7.

3.6 Radiographic inspection. The delay column shall not be missing or be short in length. There shall be no void or low density area in the area of individual pellets in the delay column. There shall be no evidence of aluminum slivers in the delay column. The flash or igniter composition shall not be missing. The boron potassium nitrate charge shall not be missing or contain voids greater than 1/16 square inch maximum. The delay washer shall not be missing. Radiographic inspection shall be conducted in accordance with 4.4.3.3.

3.7 Performance test. The following subparagraphs describe the requirements for the performance testing associated with this item. The pilot composition test phase is performed to determine the correct amount of tungsten fuel to be used to achieve the requirements as well as tests for mean time, standard deviation, and outliers using static spin stand testing and ballistic testing. The delay composition test ballistically qualifies the production batch prior to production assembly and establishes a baseline for spin stand time testing, also checking for outliers. The delay assembly test for mean time and standard deviation is performed on a sample of the loaded inspection lot, using static spin stand testing and ballistic testing as required.

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3.7.1 Pilot composition test. The pilot composition test shall be acceptable when the ballistic delay burn times of the pilot composition batches show that the required burn time of 15.4 +/- 1.0 seconds at 70 degrees F is met by at least one batch or is bracketed by at least one batch on the high and low ends so that an optimal burn time fuel ratio may be determined when tested in accordance with 4.4.3.8.1. The standard deviation of the batch(es) meeting the requirement shall not exceed .62 seconds when ballistically tested at 70 degrees F in accordance with 4.4.3.8.1. When ballistically tested at -50 degrees F, the delay shall function to the extent that it ignites the rocket motor. During static spin testing, the samples shall not contain any statistical outliers with times longer than the mean time when tested in accordance with paragraph (see 6.11). The results shall be used to set the fuel level for production.

3.7.1.1 Fuel The exact fuel/oxidizer ratios for the pilot batches shall be determined by the Government.

3.7.2 Delay composition test. Each lot of delay composition shall be ballistically tested at 70 degrees F and -50 degrees F. The production batch shall have a mean time of 15.4 seconds +/- 1 second and a standard deviation no greater than .62 seconds when ballistically tested at 70 degrees F. For -50 degree F testing, the delays shall function properly to the extent it ignites the rocket motor. Static spin stand testing shall be performed to determine suitability for ballistic testing and to establish a baseline for delay assembly testing. During static spin testing, the standard deviation shall not be greater than 0.62 seconds, and no outliers longer than the mean shall be present when tested in accordance with 4.5.7.

3.7.3 Delay assembly testing. Each production lot produced shall have an acceptable static spin mean time per 4.5.8 and 4.5.9 or an acceptable ballistic mean time of 15.4 seconds +/- 1 second at 70 degrees F and proper function at -50 degrees when tested ballistically. Ballistic testing shall only be performed with the permission of the Government.

3.8 Selector cap. After radiographic inspection of the delay the selector cap shall be assembled hand tight.

3.9 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

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3.10 Process control for delay increment consolidation. A process control description shall be developed for delay increment consolidation, and submitted for Government approval. The description shall include all elements of press setup, operation, calibration, maintenance and operator instruction required to assure that the required number of delay increments are individually consolidated at the required pressure and for the required minimum consolidation time. The process controls shall include verification that the required pressure and dwell are met at the extremes of the press consolidation range (See 4.4.3.1 and 6.10).

3.11 Particle size.

3.11.1 Tungsten particle size. The tungsten shall meet all requirements of MIL-T-48140 for TYPE I (Agglomerated) with a bulk density ranging from 50-65 grams per cubic inch. Sampling shall be as specified in 4.4.3.9 and testing shall be as specified in 4.5.6.

3.11.2 Potassium perchlorate particle size. In addition to the requirements of MIL-P-217, the particle size at the 50% level shall be 85 +/- 10 microns. Sampling shall be as specified in 4.4.3.9 and testing shall be as specified in 4.5.6.

3.12 Workmanship. The requirements for workmanship are as shown on the applicable drawing and in the applicable specification and the following:

3.12.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications required thereon. Where cleaning and preparation are required by assembly details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.12.2 Foreign matter. Parts or assemblies shall not contain dirt, grease, chips, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General Provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Delay Assembly	12926809	30
Delay Housings/Crimp Test	12929177	5/machine
Delay Assembly/For Ballistic Test	12926809	75
Delay Assembly/For Static Spin Test	12926809	75

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.2.1 Submission. One hundred fifty five (minimum quantity) delay assemblies shall be required for the first article test. Of that quantity, one hundred fifty delays shall be required for pilot composition testing in which seventy five delays shall be tested using a static spin stand test and the remaining seventy-five delay assemblies shall be packed in the format that will be used for shipping of the regular production quantity and submitted to a Government installation designated by the Contracting Officer for loading and Ballistic Testing. Five delays shall be utilized for crimp strength testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for the inspection in accordance with 4.3.1.1.

4.3.2.1.1 Testing. Proving Ground tests shall be performed on 75 assemblies for compliance with 3.7.1 utilizing the procedures of 4.5.1.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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Table I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70526A (AR)

PARAGRAPH	TITLE Delay assembly and components.	SHEET 1 OF 1	DRAWING NUMBER See below.
			NEXT HIGHER ASSEMBLY Not applicable

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Delay assembly. (Drawing 12926809)			
	Examination for defects	30	3.2	4.4.2.3, 4.4.2.4, 4.4.2.5
	Porosity test	150	3.5	4.4.3.7
	Radiographic inspection for proper loading	150	3.6	4.4.3.3
	Delay housing crimp test	5/machine	3.4	4.4.3.6
	Pilot composition test	150	3.7.1	4.5.1

NOTES

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded delay assemblies shall contain:

a. Inert parts from one interfix lot number from one manufacturer.

b. Flash composition from not more than one inspection lot.

c. Delay composition from not more than one inspection lot.

d. Igniter composition from not more than one inspection lot.

e. Boron Potassium Nitrate pellets from not more than one interfix lot.

4.4.1.1 Delay composition lot. A delay composition lot is defined as that quantity of delay material from which representative samples have been selected, loaded and forwarded to a proving ground for acceptance. A delay composition lot may be formed by any of the following methods:

a. Blending two or more batches of delay powder together to form a delay composition lot.

b. Designating a batch of delay powder or delay pellets as delay composition lot. Once samples are selected from the batch or delay composition lot, the batch or lot shall not be subsequently blended.

4.4.2 Examinations and tests.

a. Classification of Characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified herein. Attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels cited in the Classification of Characteristics paragraphs.

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[illegible] maximum resultant [illegible]

TABLE II Attributes sampling inspection

Lot Size	Inspection Level I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10

Numbers under inspection levels indicate sample size; asterisk indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative inspection provisions. Unless otherwise specified herein or provided for in the contract, alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70526A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Delay Assembly, Prior to loading delay composition.	1 OF 1	12926809
			NEXT HIGHER ASSEMBLY: 9381147/12926808

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Flash composition on sidewall	Level III	3.2	Visual
Minor 201	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

AMSMC Form 1570a-E, 1 Jul 89

Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Delay Assembly, Prior to loading igniter composition	1 OF 1	12926809
			NEXT HIGHER ASSEMBLY: 9381147/12926808

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Distance from end of housing to delay composition	Level III	3.2	Gage
Minor 201	Evidence of poor workmanship	Level V	3.12	Visual

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Delay Assembly, After crimping washer			12926809
				NEXT HIGHER ASSEMBLY
				9381147/12926808
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Washer missing	100%	3.2	Visual
102	Crimp missing or incomplete	100%	3.2	Visual
103	Depth of washer, min.	Level III	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70526A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Delay Assembly, Prior to assembly of delay closure	1 OF 1	12926809
			NEXT HIGHER ASSEMBLY: 9381147/12926808

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Boron potassium nitrate (BKNO3) charge missing	100%	3.2	Visual
102	Weight of BKNO3 charge	Level III	3.2	Balance/scale
Minor				
201	Evidence of poor workmanship	Level V	3.12	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70526A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Delay Assembly, Prior to assembly of selector cap	1 OF 1	12926809
			NEXT HIGHER ASSEMBLY: 9381147/12926808

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Gasket missing in cap	100%	3.2	Test
Major				
101	Delay closure missing or improperly positioned.	100%	3.2	Visual
102	Crimp on tube missing or body cracked by crimping operation	100%	3.2	Visual
103	Outside diameter of tube, max.	Level III	3.2	Gage
104	"O"-ring missing	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.12	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-P-70526A (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.6	Delay Assembly	SHEET 1 OF 1	12926809
			NEXT HIGHER ASSEMBLY
			9381147/12926808

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Selector cap missing	Level III	3.8	Visual
102	Identification marking missing or incorrect	Level III	3.2	Visual
Minor				
201	Evidence of poor workmanship	Level V	3.12	Visual

NOTES

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4.3 Testing.

4.4.3.1 Consolidation load. Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. Proper calibration of each consolidation machine shall be verified at the beginning, middle and end of each production shift. Dwell time and consolidation load shall be measured with a stopwatch and a load cell. If any consolidation machine fails verification for dwell time, consolidation load or for not maintaining proper load through the dwell time, all the delays manufactured on that machine since the last acceptable verification shall be rejected. In addition, a warning device indicating failure of dwell time and failure of consolidation load shall be assembled to each consolidation machine. The warning device or mechanism shall provide an easily detected signal to the machine operator. If during loading of the delay the consolidation load or dwell time requirements are not met, that delay shall be rejected.

4.4.3.2 Total volatiles. Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified total volatiles requirement. For verification, a sample of delay, igniter and flash composition from each 8 hours production and from each new container shall be selected to perform a total volatiles analysis. If any sample fails to meet the requirements of 3.3.1, the 8 hours production represented by the sample or the new container of material shall be rejected. Drying of rejected material will be accomplished with specific instructions from the Government procuring activity. The sample shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.4.3.3 Radiographic test for proper loading. Major defect. Loaded delay assemblies shall be inspected in accordance with 4.5.2 on a 100 percent basis. Any assembly not meeting the requirements of 3.6 shall be rejected and removed from the lot.

4.4.3.4 Determination of average particle size.

4.4.3.4.1 Molybdenum trioxide. Major defect. A sample shall be selected from each lot of molybdenum trioxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 9212470 for particle size, the lot shall be rejected.

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4.4.3.4.2 Chromium oxide. Major defect. A sample shall be selected from each lot of chromium oxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 9212469 for particle size, the lot shall be rejected.

4.4.3.5 Vinyl alcohol acetate resin (VAAR) solids determination. The concentration of solids in VAAR solution may change during storage due to evaporation of solvent. VAAR solids content in the delay blend can only be obtained if the solution to solids weight ratio is known prior to mixing. Prior to blending the delay composition, a sample of VAAR solution shall be tested to determine this ratio. The sample shall be tested in accordance with and meet the requirements of MIL-V-50433. Should the solids content be too high, additional solvent may be added, with agitation, until the solids content is acceptable. The VAAR solution will then be retested to assure compliance with MIL-V-50433.

4.4.3.6 Crimp strength. Major defect. At the beginning of each production shift or portion thereof, an inert delay housing shall be crimped on each crimping device using the same technique as in regular production. After crimping, the delay housing shall be sectioned perpendicular to the longitudinal axis and the load applied axially to crimp from the sectioned end. A tool manufactured to the same diameter as the delay column bore, minimum, shall be used in applying load to the crimp. If the crimp fails to meet the applicable load requirements, the crimping device shall be adjusted until the crimp meets the requirements of 3.4. The process shall then be monitored to assure that proper crimp strength is maintained throughout production.

4.4.3.7 Porosity test. This test shall be performed 100% until 1500 consecutive units have passed without failure. After the above criteria has been met, sampling shall be performed randomly at the rate of 18 delay assemblies for each shift production or portion thereof. If any one of the 18 delay assemblies fail to meet the requirements of 3.5, the total shift production shall undergo the porosity test and an additional 1500 units tested to quality sampling again.

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4.4.3.8 Performance testing.

4.4.3.8.1 Pilot composition test. For first article and each new lot of constituent ingredients, a pilot composition test shall be performed on 150 delay assemblies utilizing both ballistic and static spin testing. Five pilot batches are to be produced at tungsten contents differing from sample to sample by at least one-half part by weight. From each of the five completed batches of delay composition, enough material shall be selected to load 30 delay assemblies. If the sample is taken before the entire batch is pelletized, the representative sample shall be selected in 30 approximately equal increments taken from different areas in the mix, tumbling the mix often so as to minimize the chance of misleading results from segregation. If the sample is taken from a pelletized batch, a random sample of pellets shall be selected for use in this test. All thirty delays from each batch shall be loaded with igniter, delay composition, flash, and have the washer installed. Fifteen of the delays shall then be selected at random for static spin testing and the remaining delays shall be loaded with the boron potassium nitrate and delay closure for ballistic testing. All delays shall have the selector caps installed. Selector caps used for spin testing may be reused only for other spin test delays provided care is taken to assure pyrotechnic material is removed from the cap and gasket prior to reuse. All thirty delays from each lot shall be porosity tested and radiographically inspected per the applicable paragraphs in this specification. Fifteen delay assemblies shall be statically spin tested in accordance with 4.5.8, and the results tested for outliers per 4.5.7. If the spin test results for the batch exhibit no long time outliers per 4.5.7, the remaining fifteen completely loaded delay assemblies shall be submitted to the Government for ballistic testing in accordance with 4.5.1. If the requirements of 3.7.1 are not met, or if the delay fails to function, the batch shall be considered to have a Major defect and shall not be used for future production without approval from the Technical Agency. If, during ballistic testing, the delay ignites the motor grain while in the gun tube, the batch shall be considered to have a Critical defect, and the testing shall cease and not resume without approval from the Technical Agency. A minimum of 8 radar delay times shall be required to determine delay time acceptance for each batch during the 70 degree F temperature phase. All delays shall function properly when ballistically tested at -50 degrees F.

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4.4.3.8.2 Delay composition lot test. From each production lot of delay composition, a random sample shall be selected and loaded into 30 sample delay assemblies. Fifteen sample delays shall be statically spin tested to determine mean time, standard deviation and the presence of long time outliers per 4.5.7. If the standard deviation requirement is met and no outliers are detected, 15 complete delay assemblies shall be produced and forwarded for ballistic testing per 4.5.1. If the sample fails to meet the requirements of 3.7.2 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance.

4.4.3.8.3 Delay assembly test. From each lot of delay assemblies, 15 samples shall be randomly selected and statically spin tested. Using the spin test baseline data generated from 4.4.3.8.2, a determination of acceptability shall be made per 4.5.9. If the spin test results are not acceptable, then fifteen completed delay assemblies shall be selected at random for ballistic testing per 4.5.1 (upon approval from the Government). If the sample fails to meet the requirements of 3.7.3 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Major defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance.

4.4.3.9 Particle size. Prior to mixing of each composition batch, or pilot mixture, a sample of tungsten and of potassium perchlorate shall be tested for particle size from each shipping or storage container from which chemicals will be used in the mix. If the particle size of any sample tested fails to comply with the requirements of 3.11, chemical from that container shall not be used in the mix. Chemicals may be retested for use after application of suitable rework. A sample of 15g minimum shall be taken from each individual container in such a manner that the sample is representative of the entire contents of the container. No lab milling, mulling, grinding or similar operation is permitted during sampling or preparation thereof.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract, see section 6 of MIL-A-48078 and 6.12 herein.

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4.5.1 Ballistic test. Fifteen delay assemblies from each delay composition batch shall be forwarded to a facility designated by the contracting officer for loading into 105MM M913 or XM927 cartridges for ballistic testing at a Government Proving Ground. To assure a proper seal the loaded delay assemblies shall be radiographed after pressing into the motor body. A fuze and selector cap shall be assembled to the test vehicle prior to temperature conditioning. Ten rounds, including propelling charges, shall be conditioned at 70 degrees F +/- 2.5 degrees F for a minimum of 24 hours. Five rounds shall be conditioned to -50 -5 degrees F for a minimum of 24 hours. The selector cap shall be removed and the round fired from a M119 Howitzer with a minimum of 25 percent tube life remaining. A M229 or M67 propelling charge shall be used in these tests. At least two smear cameras shall be mounted near the gun muzzle to photograph the projectile in flight. One camera will provide a side view to assure metal parts integrity. Another camera will provide a rear view to determine presence and ignition of the delay assembly. Delay time shall be determined by use of radar. Stopwatch times may be used as a back-up to confirm that the delay functioned but shall not be used in computations for delay acceptance.

4.5.2 Radiographic test for proper loading. All delays shall be subjected to radiographic examination. In all views the radiographic beam shall be perpendicular to the longitudinal axis of the delay assembly. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks or other extraneous features should project into the region of inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-00453 (see 6.8). In lieu of a penitrometer, a radiographic standard shall be manufactured to assure adequate sensitive of the radiographic technique. A delay will be manufactured with a low density pocket in the column. Normal production materials shall be used to create the standard delay. In the column the flash and first delay increment shall be pressed in correctly. The second and third delay increments shall be placed in the column but not pressed, the fourth to last increments will be pressed. The column will be back drilled, have igniter pressed in and have the washer assembled. The delay manufactured as above will have a void between the first and second delay increments. The void must be clearly visible when the standard delay is radiographed. Using the standard delay, film sensitivity shall be checked on each piece of film used for delay acceptance. Photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.4.

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4.5.3 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive train. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. (Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units). Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within 0.5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by the Technical Agency. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by 0.05 or more shall be replaced with a new, unused step tablet.

4.5.4 Operator qualification. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II.

4.5.5 Porosity test. (See 6.5) Each delay assembly shall be conditioned at plus 145 F +/- 5 F and assembled to the porosity test equipment. Nitrogen gas under a pressure of 1000 PSIG plus or minus 50 PSIG shall be applied to the delay column at the threaded end of the housing. The pressure at the exit end of the delay column shall be recorded at 0.5 seconds, 80 seconds and 110 seconds. The pressure reading at 0.5 seconds shall be recorded and used as the zero reference.

4.5.5.1 Porosity test equipment. The contractor shall design and construct a porosity test fixture capable of testing delays to the parameters of 3.5 and 4.5.5. M549 drawings 9246401 and 9246420 may be used as a guide.

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4.5.6 Particle size.

4.5.6.1 Particle size analyzer. Sedigraph particle analyzer 5000D or an approved equivalent instrument that uses a finely collimated beam to measure particle concentration as a function of time is required.

4.5.6.1.1 Dispersing fluid. The sedimentation fluid shall be Sedisperse A12, a product of Micromeritics, or an approved equivalent.

4.5.6.2 Sedigraph test sample. Fill a 25 Ml beaker with dispersing fluid and adequate sample powder for 100% recorder pen deflection when the beaker is placed in test position and the instrument is running and has been adjusted to section 7-12 of ASTM-B430.

4.5.6.2.1 Carrier density/viscosity. If sedisperse A12 is used as a carrier, the carrier density and viscosity is supplied on the bottle label by Micromeritics for one degree temperature changes in a range of 24-36 degrees centigrade. If another carrier is used, then the user must establish these temperature dependent physical properties, before measurements.

4.5.6.2.2 Sedimentation rates. Sedimentation rates may be calculated using procedure outlined in MIL-M-382 under 4.5.11.2 with these exceptions:

a. P. = density of sedisperse A12 or another compatible fluid.

b. n. = viscosity of sedisperse A12 or another compatible fluid.

4.5.7 Dixon test for statistical outliers. This test is to be performed on static spin delay times only. For determining a long-time outlier at the 95% level of confidence, order the delay burn times from longest to shortest (T1-T15 with T1 being the longest). Calculate the Dixon ratio (R) by the following:

$$R = \frac{(T3-T1)}{(T13-T1)}$$ (Note that T15 is the shortest time)

If R is greater than .525, then the longest time is an outlier and the batch shall be rejected.

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4.5.8 Static spin stand testing. The fifteen delay assemblies that shall be furnished for this test will be comprised of a delay housing with the flash, igniter, delay composition, washer, and selector cap assembled. The delay assemblies shall be porosity tested and radiographically examined prior to spin testing. The apparatus for spin stand testing shall be comprised of a stand suitable for safe ignition of the delay column that has a fixture that shall hold the delay assembly, without slippage, while it is rotating along its longitudinal axis at 20,000 RPM. An electronic squib fixtured near the washer opening shall be used to initiate the igniter after the fixtured delay has stabilized at 20,000 +/- 200 RPM. An electronic timer triggered by the squib ignition and terminated by the flash charge output shall display total burn time. An tachometer shall be used to verify delay spin. The temperature of the delay containment fixture and delay shall be 75 +/- 15 degrees F before performing each spin test. Delay burn time shall be recorded to the nearest 0.1 second and the average delay burn time and standard deviation shall be calculated for each batch tested. The recorded data shall be used for determination of long time outlier per 4.5.7.

4.5.9 Comparison test for production batch acceptance. The difference between ballistic flight and static spin mean times from the delay composition test shall be added to the static spin mean time of the lot being tested. If the calculated mean time falls within the requirement, the lot shall be accepted.

5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Code of Federal Regulations(CFR), Title 49, Parts 100-1099. Appropriate cushioning and partitions shall be provided to protect the item from physical damage. Sufficient fillers must be used to ensure tight pack. In addition, adequate protection to damage from moisture must be provided. (Enclosing a number of items in a sealed barrier bag, meeting the requirements of Type I or Type II, Class E, Specification MIL-B-117 would satisfy this requirement.)

5.2 Marking. The items, packaged as in 5.1 shall be marked in accordance with Code of Federal Regulations (CFR) Title 49, Parts 100-129. The marking nomenclature shall be as follows:

IGNITERS, JET THRUST (JATO)
CLASS B, EXPLOSIVE

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6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The delay assemblies covered by this specification are intended for use on the 105MM, HE, RA, Projectiles M913 and XM927.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of the DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

6.3 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.4 Ammunition data cards. Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.

6.4.1 Submission of ballistic test data. In addition to the normal distribution of records, one copy of all ballistic data shall be forwarded to:

COMMANDER, ARDEC
ATTN: SMCAR-QAR-R
Picatinny Arsenal, NJ 07806-5000

6.5 Handling of delay assemblies. Precaution should be taken during loading, intraplant shipment for in process test, subsequent to loading and prior to packing to assure delays are not exposed to moisture. Delay assemblies should not be removed from moisture proof packing until such time as the units are to be placed in the conditioning chamber for porosity testing and flight testing.

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6.6 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.7 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.8 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, ARDEC, Picatinny Arsenal, 07806-5000, ATTN: SMCAR-QAR-R for review and approval upon implementation at each six month period of production.

6.9 Source of supply. The model 5000D Sedigraph can be secured from Micromeritics Inst. Corp.; Norcross, Georgia, USA.

6.10 Submission of process control description. The process control description for delay increment consolidation shall be prepared in accordance with DIP-1604 and submitted to the responsible Government technical agency for approval.

6.11 Static spin versus ballistic delay burn times. Testing has shown that delay assemblies burn approximately 0.75 to 1.25 seconds faster on the static spin stand than in ballistic testing.

6.12 Submission of inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.13 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

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6.14 Subject term (key word) listing.
Pyrotechnic delay

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1315-0022)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70526 (AR)	940706

3. DOCUMENT TITLE: PROJECTILE, 105MM, HE, RA, M913/XM927 DELAY ASSEMBLY FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. MAILING ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial: 201-724-6675 (2) AUTOVON: DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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